
Unit – 3:

OS Security: Introduction: Secure OS, Security Goals, Trust Model, Threat Model, Access Control. Fundamentals: Protection system, Lampson's Access Matrix, Mandatory protection system.

1. OS SECURITY – INTRODUCTION

Definition

Operating System Security (OS Security) refers to the set of mechanisms, policies, and tools used to protect the operating system from unauthorized access, misuse, modification, or destruction. It ensures the **Confidentiality, Integrity, and Availability (CIA)** of OS resources.

★ 2. SECURE OS (Secure Operating System)

Definition

A **Secure OS** is an operating system that is designed with built-in security mechanisms to prevent attacks, enforce policies, and ensure only authorized access.

Characteristics of a Secure OS

1. **Strong authentication**
2. **Least privilege principle**
3. **Process isolation**
4. **Secure memory management**
5. **Access control enforcement (MAC & DAC)**
6. **Audit logs**
7. **Secure file system**
8. **Protection from malware**

Examples

- SELinux
 - Windows Server with hardened configuration
 - Trusted Solaris
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★ 3. SECURITY GOALS (CIA TRIAD)

OS security aims to achieve the following:

1. Confidentiality

- Prevent unauthorized access to data
- Techniques: Encryption, access control, password protection

2. Integrity

- Prevent unauthorized modification of data
- Example: Digital signatures, checksums

3. Availability

- Ensure system resources are accessible when needed
- Example: Backups, DoS protection

Additional Goals (sometimes part of exam)

- **Authenticity** – verifying identity
 - **Accountability** – tracking user actions with logs
 - **Non-repudiation** – users cannot deny their actions
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★ 4. TRUST MODEL

Definition

A **trust model** defines *who/what is trusted* in the system and how that trust is established, maintained, and revoked.

In OS security:

Trusted Components

- Kernel
- System processes
- Authentication mechanisms
- Security modules (e.g., SELinux policies)

Key Points

- Trust must be **minimized** (least trust principle)
 - Trusted components must be **verified and protected**
 - The OS relies on a **Trusted Computing Base (TCB)**
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★ 5. THREAT MODEL

Definition

A **threat model** identifies *potential attackers*, their *capabilities*, *behaviors*, and *possible attack paths*.

Threat Model Includes

1. Threat Actors

- Hackers
- Malicious insiders
- Malware
- External attackers

2. Threat Vectors

- Network attacks
- Malware infection
- Physical access
- Exploiting OS vulnerabilities

3. Threat Scenarios

- Privilege escalation
- Data corruption
- Unauthorized file access
- DoS attacks

Purpose

To design OS security controls to reduce risks.

[Important Difference :](#)

Criteria	Trust Model	Threat Model
Definition	A security framework that defines which entities are trusted , the level of trust , and conditions of trust .	A structured process that identifies potential threats, attackers , and ways the system can be attacked .
Main Focus	Focuses on trust relationships between users, systems, or components.	Focuses on identifying risks, attack vectors, and vulnerabilities .
Purpose	To establish who/what can be trusted in a computing environment.	To understand who can attack the system and how attacks may occur .

Criteria	Trust Model	Threat Model
Type of Assumption	Makes positive assumptions (some entities are reliable).	Makes negative assumptions (any entity can be a potential attacker).
Concerned With	Authentication, authorization, certificates, trusted computing base (TCB).	Attackers, threats, vulnerabilities, risk impact, attack surface.
Key Question Answered	“Whom should we trust?”	“What can go wrong? Who can harm us?”
Orientation	Constructive model – enables secure operations based on trust.	Defensive model – prepares protection mechanisms based on threats.
Used In	PKI, certificate chains, OS trust boundaries, secure boot, TPM.	STRIDE, DREAD, risk assessment, cybersecurity planning.
Output	A set of trusted entities and the trust boundaries .	A list of threats, risks, and mitigation strategies .
Viewpoint	Looks at the internal reliability of the system.	Looks at external/internal attackers and possible compromises .
Example	Users with admin privileges, trusted kernel modules, trusted network nodes.	Malware threats, DDoS attacks, MITM attacks, insider threats.
Goal	To build a secure foundation of trust .	To identify and minimize risks .

★ 6. ACCESS CONTROL (Basics)

Definition

Access Control refers to **mechanisms that control who is allowed to access which resources** and what actions they can perform.

Types

1. **DAC (Discretionary Access Control)**
 - User decides permissions
 - Example: File permissions in Windows/Linux
2. **MAC (Mandatory Access Control)**
 - System enforces strict policies

- Used in secure OS (military)

3. RBAC (Role-Based Access Control)

- Permissions based on roles
- Example: Admin, User, Guest

Elements

- Subjects → Users, processes
 - Objects → Files, devices
 - Access rights → Read, write, execute
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★ 7. FUNDAMENTALS OF PROTECTION SYSTEM

Definition

A **protection system** defines how an OS protects resources from unauthorized access. It enforces rules that specify **who can do what**.

Goals

- Correctness (rules implemented properly)
- Completeness (covers all resources)
- Efficiency (fast and practical)

Components

1. **Objects** – resources to protect (files, memory, devices)
2. **Subjects** – users or processes
3. **Access Rights** – read, write, execute, delete

Protection systems ensure:

- Isolation of processes
 - Safe sharing of resources
 - Prevention of interference
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★ 8. LAMPSON'S ACCESS MATRIX

Definition

Lampson's Access Matrix is a **formal model** describing how access control is represented in an OS.

It shows **which subject** has **what rights** over **which object**.

Structure

It is a **two-dimensional matrix**:

Subject / Object File 1 File 2 Printer Memory

User A Read Write — Execute

User B — Read Print —

Elements

1. **Rows = Subjects**
 2. **Columns = Objects**
 3. **Matrix Cells = Access Rights**
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Advantages

- Clear representation
 - Easy to implement in ACL-based systems
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Implementations

1. **Access Control Lists (ACL)**
 2. **Capability Lists (C-Lists)**
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★ 9. MANDATORY PROTECTION SYSTEM (Mandatory Access Control – MAC)

Definition

A **mandatory protection system**, or Mandatory Access Control (MAC), is a strict access control model where the **system**, not the user, decides access permissions.

Used in **military, defense, critical security systems**.

Key Characteristics

1. **Security labels and classifications**
 - Example: Top Secret, Secret, Confidential, Unclassified
2. **Subjects and objects have security levels**
3. **Users cannot change permissions**
 - OS controls everything

4. Rules based on Bell–LaPadula and Biba models

Policies

Bell–LaPadula (Confidentiality Model)

- **No Read Up** (simple security rule)
- **No Write Down** (star rule)

Biba (Integrity Model)

- **No Read Down**
 - **No Write Up**
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Advantages

- Strong security
- Prevents data leakage
- Highly controlled environment

Disadvantages

- Less flexible
 - Difficult to manage
 - Not suitable for every organization
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★ FINAL QUICK SUMMARY (FOR REVISION)

Topic	Meaning
Secure OS	OS designed with built-in security (authentication, isolation, ACLs)
Security Goals	CIA: Confidentiality, Integrity, Availability
Trust Model	Defines what is trusted in OS (kernel, TCB)
Threat Model	Identifies attackers and possible attack paths
Access Control	Mechanisms controlling user access (DAC, MAC, RBAC)
Protection System	Rules for protecting resources
Lampson's Access Matrix	Matrix showing which subject has what access
Mandatory Protection System	Strict OS control with security labels (MAC)

Topic

Meaning